

Vaibhav Talekar

AI Specialist & Machine Learning Engineer

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LINKS	LinkedIn , GitHub
PROFILE	Results-driven AI Specialist and Machine Learning Engineer with 3+ years of experience delivering production AI systems and commercial full-stack applications. Currently pursuing MSc Artificial Intelligence at University of Surrey while building scalable automation solutions for business clients. Proven track record of leading cross-functional teams and delivering cutting-edge AI solutions from research to production. Strong expertise in transformers, ensemble methods, end-to-end ML pipelines, and modern web technologies (Next.js, React, TypeScript).

EMPLOYMENT HISTORY

Nov 2025 — Present	Freelance Software Developer & AI Automation Specialist <i>Guildford</i> <i>Independent Consultant - Business Automation & AI Solutions</i> - Developed and sold commercial full-stack applications to multiple business clients, generating revenue through custom automation solutions and ERP systems - Built enterprise ERP system for PMR Industries managing multi-warehouse inventory (3 warehouses, 14+ product types), production workflows (Urea → DEF conversion), financial tracking with 5 account types, and automated Google Drive backups - Architected Solar Sales Management System with 15-step workflow tracker, role-based authentication, RESTful APIs using Supabase (PostgreSQL), Cloudflare R2/AWS S3 integration, and automated PDF generation - Created AI-driven business outreach platform with custom prompt engineering for automated email generation, third-party API integration (Smartlead, OpenAI), real-time streaming search, and Vercel Blob storage - Tech Stack: Next.js 15, TypeScript, React, PostgreSQL, Prisma, Supabase, Node.js, Tailwind CSS, Recharts, AWS S3, Cloudflare R2, OpenAI API
Feb 2025 — Nov 2025	Software Developer (AI Specialist) , zzish (Marvely) <i>Guildford</i> <i>EdTech Company - AI-powered language learning platform</i> - Led AI backend development for multi-language speaking practice platform serving GCSE, IGCSE, and IELTS preparation - Architected speech generation, feedback systems, and automated reporting using advanced NLP models - Built complete V2 platform using Svelte 5, delivering 40% faster performance and enhanced error handling - Integrated payment systems and developed scalable backend infrastructure supporting thousands of users
Mar 2022 — Feb 2025	Subject Matter Expert (Data Science) , Scaler Academy <i>Bengaluru</i> - Developed computer vision modules leveraging machine learning techniques for enhanced image analysis - Designed machine learning models using complex algorithms to solve intricate problem domains - Contributed to curriculum development, integrating cutting-edge technologies to enhance student learning experience

EDUCATION

Feb 2025 — Feb 2026	Master of Science in Artificial Intelligence , University of Surrey <i>Guildford</i> Semester 1: 74.88% Average (Distinction) 120 Credits Awarded Graduation Expected: April-May 2026 Relevant Modules: Applied Machine Learning, Computer Vision, Natural Language Processing, AI Programming, Research Professionalism and Innovation, Thesis
Aug 2019 — Jul 2023	Bachelor of Technology in Information Technology , Savitribai Phule Pune University <i>Pune</i> CGPA: 9.04/10 (First Class with Distinction)

SKILLS

AI/ML	Deep Learning (Expert) • TensorFlow (Expert) • PyTorch (Experienced) • Transformers (Experienced) • Computer Vision (Expert) • Feature Engineering (Expert) • LLMs (Skillful) • NLP (Experienced)
Full-Stack	Python (Experienced) • JavaScript/TypeScript (Skillful) • Next.js (Experienced) • React (Experienced) • Svelte (Skillful) • Node.js (Experienced) • PostgreSQL (Experienced) • MongoDB (Skillful)
Tools/Cloud	AWS S3 • Cloudflare R2 • Supabase • Prisma • Vercel • Git • RESTful APIs • Docker • LangChain • OpenAI API

KEY ACADEMIC PROJECTS

Mar 2025 — Present	Protein Phosphorylation Site Prediction <i>Guildford</i> <i>(MSc Thesis Project - Research Paper Under Review with Positive Feedback)</i> - Achieved 82.8% accuracy using novel ensemble approach combining feature engineering with transformer models - Implemented ESM2 transformer (Facebook's protein language model) with optimized performance - Conducted 100+ experiments across XGBoost, deep learning, and ensemble methods - Developed comprehensive feature extraction pipeline with 5x faster processing through optimization
Apr 2025 — Jun 2025	Glaucoma Detection using Deep Learning <i>Guildford</i> <i>(Team Leader - 31% Contribution)</i> - Led team of 4 developing automated glaucoma detection achieving 94.7% Dice score and 90% IoU - Implemented multiple architectures (UNet, FPN, DeepLabV3+, SegFormer) with comparative analysis - Optimized loss functions using Dice-Focal combination addressing class imbalance - Delivered clinically relevant solution with Cup-to-Disc Ratio estimation
Apr 2025 — Jun 2025	Network Intrusion Detection System <i>Guildford</i> <i>(Academic Project - Required 90% accuracy to pass)</i> - Developed RBFN model from scratch achieving 92.37% training and 88.6% testing accuracy - Implemented comprehensive preprocessing including feature selection and standardization

	Performed hyperparameter optimization using grid search, improving accuracy to 90.58%
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ACHIEVEMENTS & RECOGNITION

Mar 2025 — Present	Research Publication (In Progress) - Protein Phosphorylation Site Prediction (Paper under review with positive feedback)
Jan 2024 — Jan 2026	DataCamp Data Scientist Professional Certificate Skills: Statistical Data Analysis • Data Visualization • Machine Learning • Deep Learning • Data Science
Jan 2022 — Mar 2022	Runner-up (1st Position) - Barclays Tech Innovation Intercollegiate Challenge

LEADERSHIP & ACTIVITIES

Nov 2022	Organizer, Code Buddies Coding Event Orchestrated planning and execution of coding competition, managing logistics and team coordination
Aug 2021 — Mar 2023	Volunteer, Training and Placement Cell Managed recruitment logistics for visiting companies, facilitating seamless placement drives

LANGUAGES

LANGUAGES	English (Highly proficient) • Hindi (Native speaker) • Marathi (Native speaker)
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